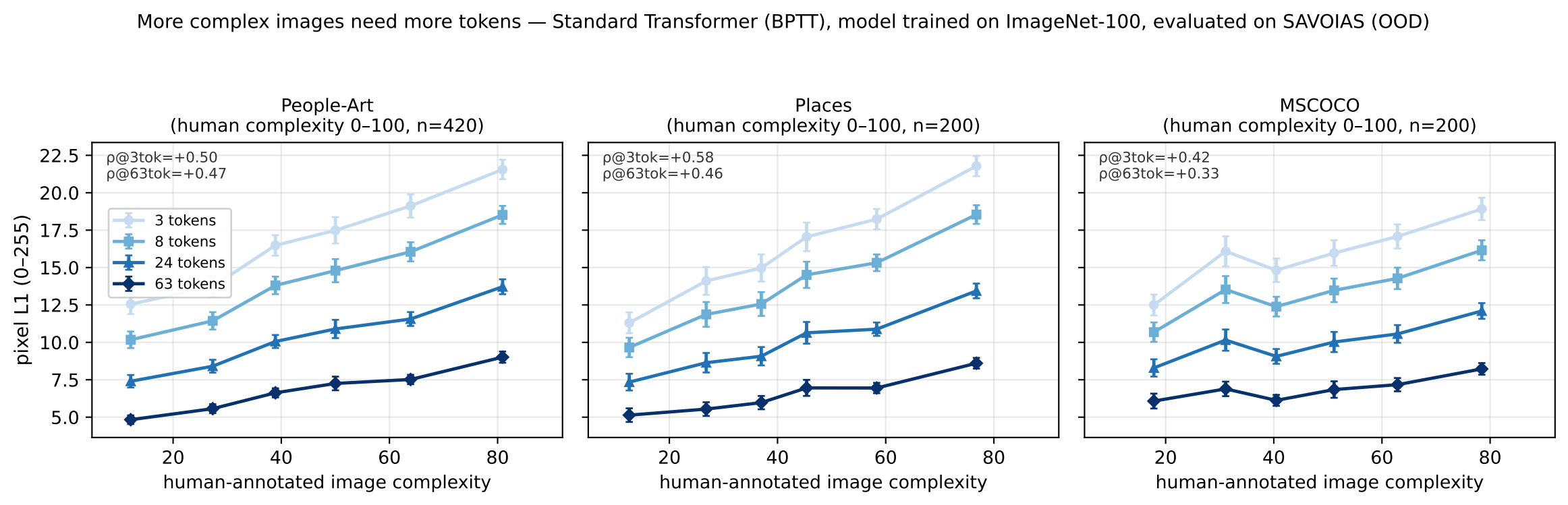
More complex images need more tokens — Standard Transformer (BPTT), model trained on ImageNet-100, evaluated on SAVOIAS (OOD)
People-Art
(human complexity 0–100, n=420)
Places
(human complexity 0–100, n=200)
MSCOCO
(human complexity 0–100, n=200)
ρ@3tok=+0.50
ρ@63tok=+0.47
ρ@3tok=+0.58
ρ@63tok=+0.46
ρ@3tok=+0.42
ρ@63tok=+0.33
3 tokens
8 tokens
24 tokens
63 tokens
pixel L1 (0–255)
22.5
20.0
17.5
15.0
12.5
10.0
7.5
5.0
20
40
60
80
human-annotated image complexity